THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
7	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
10	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
11	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
12	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
13	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
14	**(d)** 3 x 10^5	2 x 10^30 / 6 x 10^24.
15	**(d)** 50 per cent	30/60.
16	**(a)** 2 : 1	75/25 = 3; 60/40 = 1.5; ratio 2 : 1.
17	**(a)** Both A and R are true and R is the correct explanation of A.	Some readings are slightly too high and some too low, so the mean reduces the effect of random errors.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Sand stays on the filter paper while the salt solution passes through; the difference in solubility explains the method.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Since the number and kind of atoms stay the same, the total mass is conserved.
20	**(a)** Both A and R are true and R is the correct explanation of A.	Alpha particles pass through the empty space and are deflected only when they come close to the tiny nucleus.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Area = 96 square cm (1); About 2 per cent (1).	1+1
22	Mass of solution = 240 g (1); 24/240 x 100 = 10 per cent (1).	1+1
23	4 g H needs 32 g O (1); Water = 36 g; 8 g of oxygen left (1).	1+1
24	63x + 65(100 - x) = 6350 (1); x = 75, so the heavier isotope is 25 per cent (1).	1+1
25	Harmonic mean 2 x 30 x 60/90 (1); = 40 km/h (1).	1+1
26	a = -2 m/s squared (1); t = 5 s, s = 25 m (1).	1+1

SECTION — C (Short Answer) — Answers

6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Mean = 42.6/3 (1); = 14.2 cm (1); Greatest deviation = 0.1 cm (1).	1+1+1
28	320 g contains 120 g salt and 200 g water (1); At 20 degrees Celsius 200 g water holds 50 g (1); Crystals = 120 - 50 = 70 g (1).	1+1+1
29	S: 1/32 = 0.03125 (1); O: 1.6/16 = 0.1 (1); Ratio about 1 : 3: SO_3 (1).	1+1+1
30	2, 8, 3 (1); M^{3+} (1); Mass = 54 + 48 = 102 u (1).	1+1+1
31	v = 20 m/s; s1 = 50 m (1); s2 = 200 m (1); t3 = 10 s; s3 = 100 m; total = 350 m (1).	1+1+1
	OR: a = 4 m/s squared (1); F = 6000 N, s = 50 m (1); Weight = 15000 N; force is 0.4 of weight (1).	1+1+1
32	PE lost = 200 J (1); KE gained = 90 J (1); Work against friction = 110 J (1).	1+1+1
	OR: Wavelengths 4 : 1 (1); Speeds equal (1); Loudness equal; pitch higher for 400 Hz (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) 1 mm = 0.1 cm (1). (ii) $(14.2+14.4+14.3+14.1+14.5)/5 = 71.5/5 = 14.3$ cm (1). (iii) $14.3\ \text{cm} = 0.143$ m (1). (iv) Readings that are too high and too low cancel partly in the mean, so random errors are reduced (1); the largest deviation is $\lvert 14.5-14.3\rvert = \lvert 14.1-14.3\rvert = 0.2$ cm (1).	1+1+1+2
34	(i) Solute: salt; solvent: water (1). (ii) $20+180=200$ g (1). (iii) $\frac{20}{200}\times 100 = 10\%$ (1). (iv) For 5 per cent, total mass $= \frac{20}{0.05} = 400$ g (1); water to be added $= 400-200 = 200$ g (1).	1+1+1+2
35	Run 1: O = 1.6 g, Mg : O = 3 : 2 (1); Run 2: O = 4.0 g, Mg : O = 3 : 2 (1); Same ratio in both runs (1); Formula MgO: 24 : 16 = 3 : 2 (1); Law and formula verified (1).	1+1+1+1+1
36	69x + 71(100 - x) = 6980 (1); x = 60, so 60 per cent and 40 per cent (1); Neutrons: 38 and 40 (1); Element is gallium (1); Average mass between 69 and 71 (1).	1+1+1+1+1
	OR: Distances inversely proportional to a: 2 km and 1 km (1); v squared = 2 x 0.5 x 2000, v = 44.7 m/s (1); t1 = 89.4 s (1); t2 = 44.7 s; total 134 s (1); Check s2 = 44.7 squared/2 = 1000 m (1).	1+1+1+1+1
37	a = 3 m/s squared (1); v = 15 m/s (1); s = 37.5 m (1); p = 120 kg m/s (1); F = change in p/t = 24 N (1).	1+1+1+1+1
	OR: v = sqrt(2 x 10 x 3.2) = 8 m/s (1); KE at bottom = 64 J (1); Work by friction = -64 J (1); f x 4 = 64, f = 16 N (1); Energy conserved: PE to KE to heat (1).	1+1+1+1+1
38	lambda = 0.782 m (1); t = 344/344 = 1 s (1); Waves between source and wall = 172/0.782 = 220 (1); Echo path 344 m: 440 waves (1); Frequency equals the waves per second (1).	1+1+1+1+1
	OR: Cut equal potato cubes, weigh (1); Place in solutions for same time (1); Reweigh and calculate the percentage change (1); Plot the graph of percentage change vs concentration (1); Intercept on the x-axis gives isotonic concentration (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 0.68 m and 3 m	lambda = v/f.
2	**(b)** hypotonic to the cell	Water entered.
3	**(d)** accumulation of food	Phloem blocked.
4	**(c)** Fungi	Hyphae and chitin.
5	**(b)** 14 March	Mid cycle.
6	**(a)** 4000 J	400 W x 10 s.
7	**(b)** the theory is strongly supported but remains open to tests	Science is provisional.
8	**(a)** only one component sublimes	Different behaviour on heating.
9	**(a)** 4.4 g	Mass of gas escaped.
10	**(d)** 24.3 u	(1896 + 250 + 286)/100 = 24.32.
11	**(d)** 8 km/h	24 km/ 3 h.
12	**(a)** the pull of the book on the Earth	Third-law pair.
13	**(c)** 250 J	mgh.
14	**(d)** 24 m/s	0.6 x 40.
15	**(d)** synthesis of ATP	Mitochondria do ATP.
16	**(b)** glandular epithelium	Secretion.
17	**(a)** Both A and R are true and R is the correct explanation of A.	If a body returns to its starting point, the shortest distance between start and finish is zero, although it has travelled a path.
18	**(a)** Both A and R are true and R is the correct explanation of A.	This is inertia of motion, which is the reason for the forward jerk.
19	**(a)** Both A and R are true and R is the correct explanation of A.	There is no displacement, so $W = Fs = 0$.
20	**(a)** Both A and R are true and R is the correct explanation of A.	With no particles to vibrate, no compressions and rarefactions can form.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	KE change = (1/2)(2)(25 - 9) (1); = 16 J (1).	1+1
22	Distance = 170 m (1); v = 170/0.5 = 340 m/s (1).	1+1
23	SA/V = 6 and 2 (1); Larger cell exchanges less efficiently, so it divides (1).	1+1
24	Secondary xylem accumulates every year (1); Lignified walls give strength (1).	1+1
25	New evidence from molecular studies (1); Discovery of new organisms (1).	1+1
26	Muscle 46, sperm 23 (1); Egg 23, zygote 46 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	0.5 per cent: gains mass (1); 5 per cent: loses mass (1); 1 per cent is isotonic with the potato cell sap (1).	1+1+1
28	Potato turns blue-black as it stores starch (1); Carrot gives less blue-black (1); Parenchyma stores food (1).	1+1+1
29	Feathers? yes: pigeon (1); No: has hair? yes: bat (1); No: gills with scales: shark; moist skin: frog; dry scaly skin: snake (1).	1+1+1
30	Ovulation about 14 days before the next period: day 16 (1); Fertile period: days 13 to 17 (1); Sperms survive 3 days before (1).	1+1+1
31	Annual emission = 250 x 12 x 0.8 = 2400 kg (1); Trees = 2400/22 (1); = 109 trees (1).	1+1+1

	OR: Shows how reliable the result is (1); e.g. 12.3 +/- 0.1 cm (1); Allows comparison with other results (1).	1+1+1
32	A: 5/55 = 9.1 per cent (1); B: 8/108 = 7.4 per cent (1); A is more concentrated (1).	1+1+1
	OR: Mg2+ and O2- attract more strongly (1); More energy to break the lattice (1); Hence higher melting point (about 2800 vs 801 degrees Celsius) (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) In a chemical reaction the total mass of the reactants equals the total mass of the products (mass is neither created nor destroyed) (1). (ii) $10 - 5.6 = 4.4$ g (1). (iii) Antoine Lavoisier (1). (iv) No (1); carbon dioxide escapes into the air from the open dish, and if its mass is included the total mass stays 10 g (1).	1+1+1+2
34	(i) Atomic number = number of protons = 11; mass number = 11 + 12 = 23 (1). (ii) 2, 8, 1 (1). (iii) Valency = 1 (it has 1 valence electron) (1). (iv) The atom loses its one outer electron (1) to give X^{+} with 10 electrons and configuration 2, 8 (1).	1+1+1+2
35	Tube 1: moist cotton, room temperature (1); Tube 2: dry cotton (1); Tube 3: seeds submerged in boiled water with oil (1); Tube 4: moist cotton in a fridge (1); Only tube 1 germinates; conclusion (1).	1+1+1+1+1
36	Absorbed = 0.7 x 342 = 239 W/m squared (1); At equilibrium the outgoing is 239 W/m squared (1); Albedo 0.28: absorbed 246 W/m squared, warming (1); More greenhouse gases reduce the outgoing radiation (1); Surface warms until balance is restored (1).	1+1+1+1+1
	OR: Same mass and volume (1); Vary only temperature (1); Record times (1); Graph of time vs temperature (1); Errors: reaction time and stirring (1).	1+1+1+1+1
37	Particles carry charge and repel (1); Coagulation with electrolytes (1); Electrophoresis moves particles in a field (1); Dialysis separates dissolved ions (1); Examples: blood purification (1).	1+1+1+1+1
	OR: Atoms of different elements have different masses (1); 12 g of Mg has half as many atoms as 12 g of C (1); Number of atoms is proportional to mass divided by atomic mass (1); 24 g of Mg has as many atoms as 12 g of C (1); Atomic masses link mass to number of atoms (1).	1+1+1+1+1
38	n = 24 - 12 = 12 (1); B: p = 11, n = 12, A = 23 (1); B is sodium $^{23}_{11}\mathrm{Na}$ (1); 2, 8, 1 (1); A is magnesium-24 (1).	1+1+1+1+1
	OR: T = 2 x 3.14/4 = 1.57 s (1); Speed constant 4 m/s (1); Velocity changes direction (1); Displacement in half period 2 m (1); Average velocity = 2/0.785 = 2.55 m/s (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
2	**(c)** A is true but R is false.	A is true; R is false because the gas escapes.
3	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
4	**(a)** Both A and R are true and R is the correct explanation of A.	R explains A.
5	**(b)** 4 m/s squared	F/m.
6	**(a)** 480 W	2400/5.
7	**(d)** 150 m	1500 x 0.2/2.
8	**(c)** cell wall	Rigid wall.
9	**(d)** sensory neuron, relay neuron and motor neuron	Reflex arc.
10	**(a)** animal	Animalia.
11	**(c)** wings or hairs	Light structures.
12	**(a)** the ground is heated by sunlight and warms the air	Surface heating.
13	**(d)** 5.4 cm	Subtract the zero error.
14	**(a)** 32 g	Water 100 g holds 32 g; crystals 64 - 32.
15	**(c)** 3 : 8, CO_2	Atoms ratio 2.4/12 : 6.4/16 = 1 : 2.
16	**(b)** 20 and 16	X lost 2, Y gained 2.
17	**(a)** Both A and R are true and R is the correct explanation of A.	ATP is the energy currency of the cell, which is why mitochondria are called powerhouses.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Growth requires new cells, which are produced by dividing meristematic cells.
19	**(a)** Both A and R are true and R is the correct explanation of A.	Interbreeding with fertile offspring is the defining feature of a species.
20	**(a)** Both A and R are true and R is the correct explanation of A.	With one parent and no gamete fusion, the offspring are clones of the parent.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Absorbed = 0.7 x 342 (1); = 239 W/m squared (1).	1+1
22	A straight line shows proportionality (1); Easy to read the slope and spot errors (1).	1+1
23	Boiling point is fixed (1); Heat supplies latent heat (1).	1+1
24	Na+: 2, 8; F-: 2, 8 (1); Same as neon (1).	1+1
25	Maximum 2n squared with n = 1 is 2 (1); K shell is the smallest and nearest to the nucleus (1).	1+1
26	Dropped: 2 s (1); Thrown down: 20 = 10t + 5t squared gives t = 1.24 s (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	p + n = 31 and n = p + 5 (1); 2p = 26, so p = 13 and n = 18 (1); Z = 13; configuration 2, 8, 3 (1).	1+1+1
28	s1 = a/2 = 2, so a = 4 m/s squared (1); s4 = 0.5 x 4 x 16 (1); = 32 m (1).	1+1+1
29	a = v squared/2s = 2 x 10^6 m/s squared (1); F = 0.01 x 2 x 10^6 = 20000 N (1); t = v/a = 0.0002 s (1).	1+1+1
30	Energy proportional to x squared (1); Doubling the extension gives 4 times the energy (1); 16 J (1).	1+1+1

31	T = 0.00143 s (1); lambda = 0.486 m (1); Waves in 5 s = 3500 (1).	1+1+1
	OR: Break cells gently (1); Centrifuge at low speed to remove nuclei (1); Higher speed to get mitochondria (1).	1+1+1
32	Constant energy demand (1); Continuous contraction (1); Supplies ATP (1).	1+1+1
	OR: Moss: gametophyte dominant (1); Fern: sporophyte dominant (1); Both need water for fertilisation (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $v = u + at = 0 + 0.5 \times 40 = 20$ m/s (1). (ii) $s = \frac{1}{2}at^2 = \frac{1}{2} \times 0.5 \times 1600 = 400$ m (1). (iii) $s = vt = 20 \times 60 = 1200$ m (1). (iv) Total $= 400 + 1200 = 1600$ m (1); average speed $= 1600/100 = 16$ m/s (1).	1+1+1+2
34	(i) $W = mg = 20 \times 10 = 200$ N (1). (ii) $50 - 10 = 40$ N (1). (iii) $a = F/m = 40/20 = 2$ m/s^2 (1). (iv) $v = at = 2 \times 5 = 10$ m/s (1); $s = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times 25 = 25$ m (1).	1+1+1+2
35	Time gap less than 0.1 s in a small room (1); A big hall delays reflections (1); Use of absorbers (1); Curved ceilings and sound boards (1); Ceiling height above 17 m avoided or treated (1).	1+1+1+1+1
36	Stem cells unspecialised (1); Differentiation into RBC, nerve, muscle (1); Plant meristems (1); Uses: bone marrow transplants (1); Ethical issues: source of embryonic cells (1).	1+1+1+1+1
	OR: Xerophytes: thick cuticle, sunken stomata (1); Hydrophytes: aerenchyma (1); Mesophytes: normal tissues (1); Root tissues for water (1); Support tissues (1).	1+1+1+1+1
37	Roots to absorb water (1); Vascular tissue for transport (1); Seeds protect the embryo (1); Flowers for pollination (1); Fruits for dispersal (1).	1+1+1+1+1
	OR: Causes: blocked tubes, low sperm count (1); Hormonal problems (1); IVF steps (1); Artificial insemination (1); Social acceptance and ethics (1).	1+1+1+1+1
38	Less transpiration, less rainfall (1); Carbon released (1); Albedo changes (1); Biodiversity loss (1); Soil erosion and floods (1).	1+1+1+1+1
	OR: Strength: evidence-based (1); Strength: self-correcting (1); Limitation: cannot test untestable ideas (1); Chance: penicillin discovery (1); Persistence: many experiments repeated (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER